



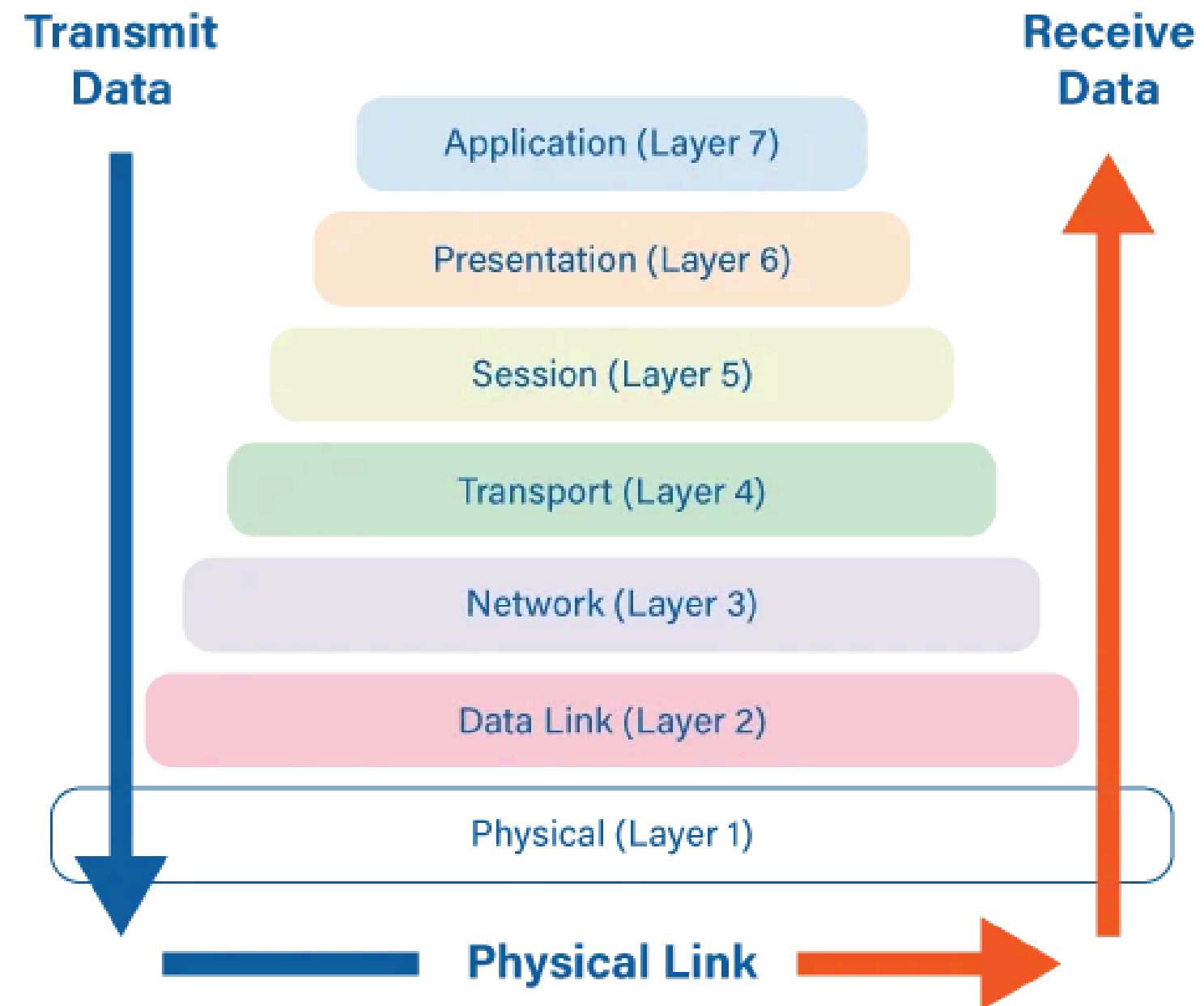
Networking Devices, Topologies and the Cloud Network

Computer Networks

Join the lecture online on your dashboard

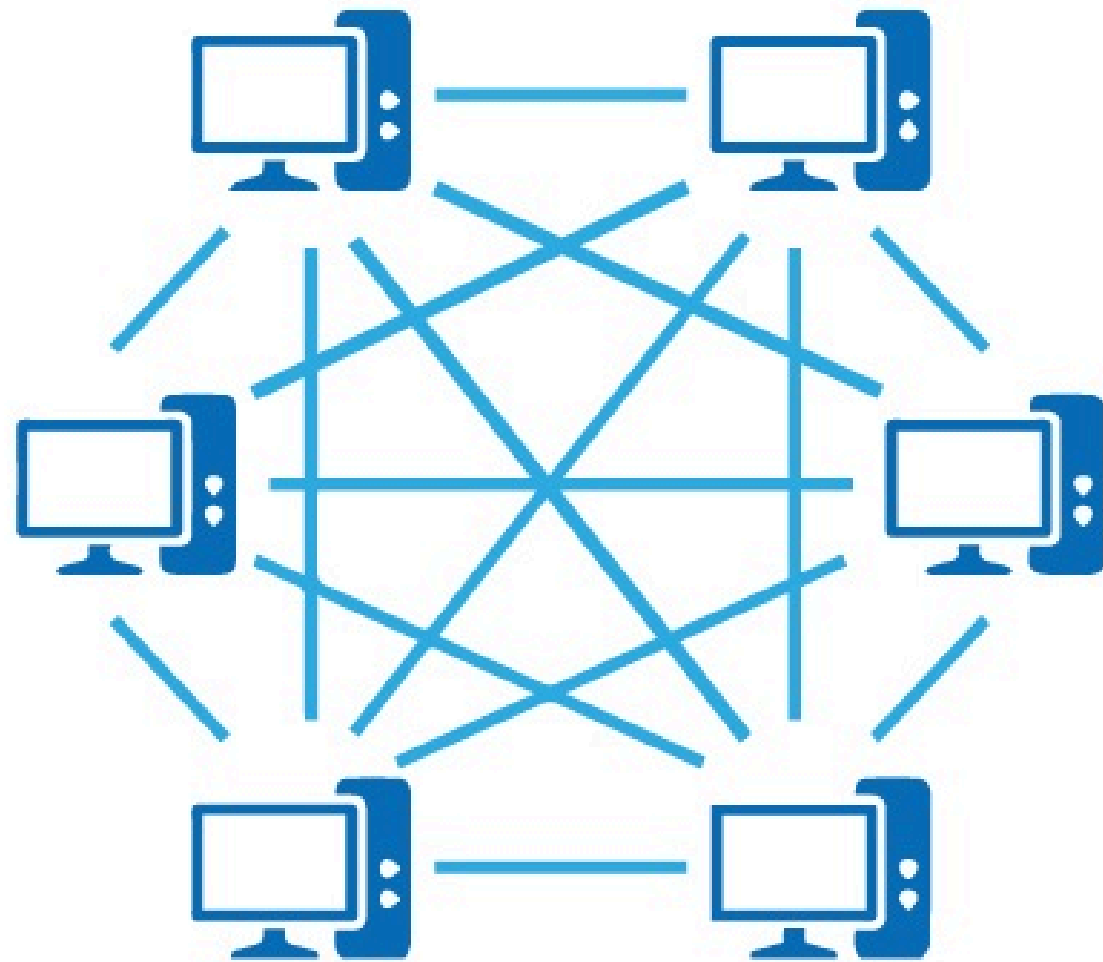
Recap

The 7 Layers of OSI

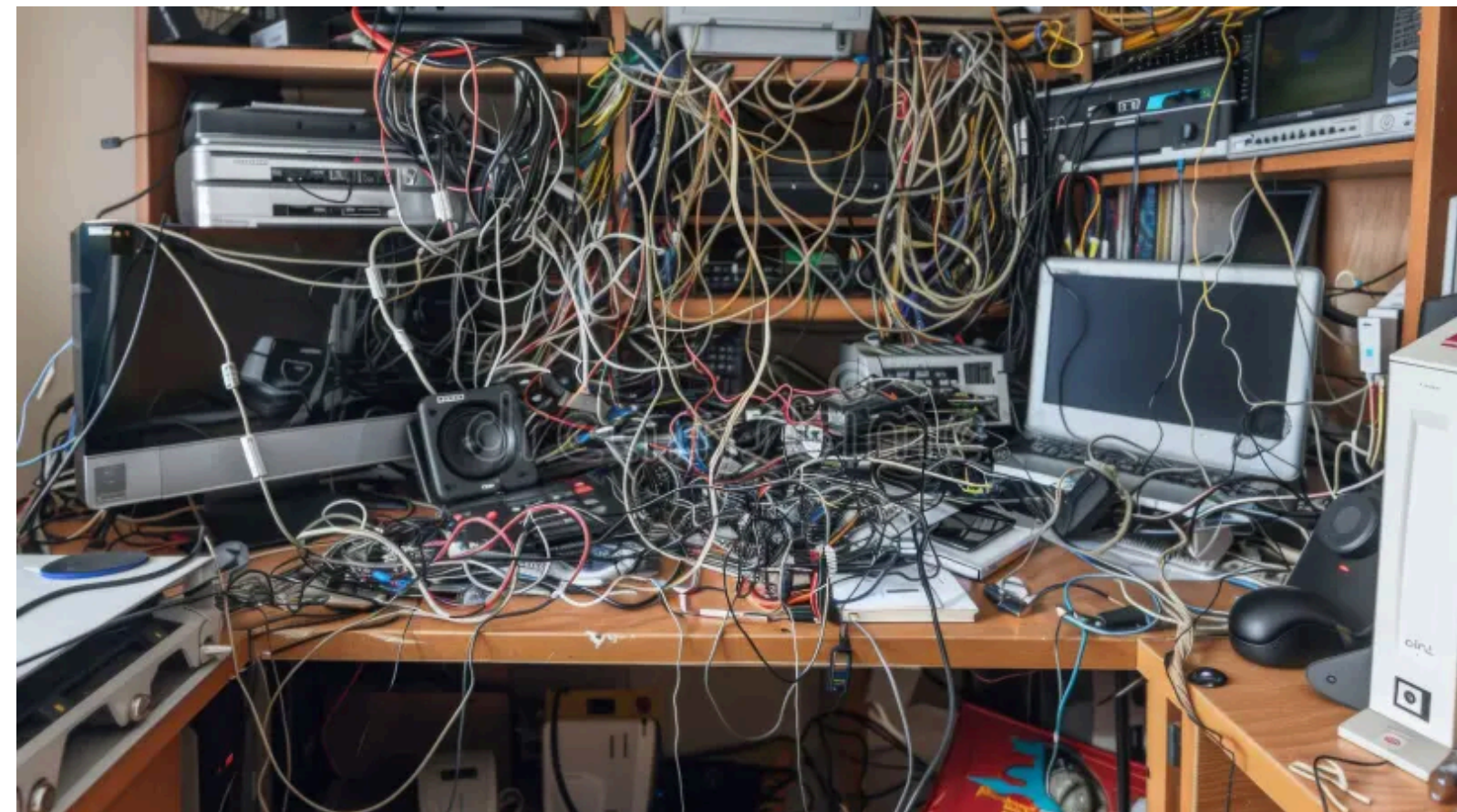


The why of network devices

Connecting computers



3 devices: $3(3-1)/2 = 3$ cables
10 devices: $10(10-1)/2 = 45$ cables



100 devices:

**Instead of connecting everyone to everyone
what if everyone connects to ONE device**

Networking Devices

- Core Components of a Computer Network
- Connect computers and other devices in a network.



Hub



Switch



Wireless
Router

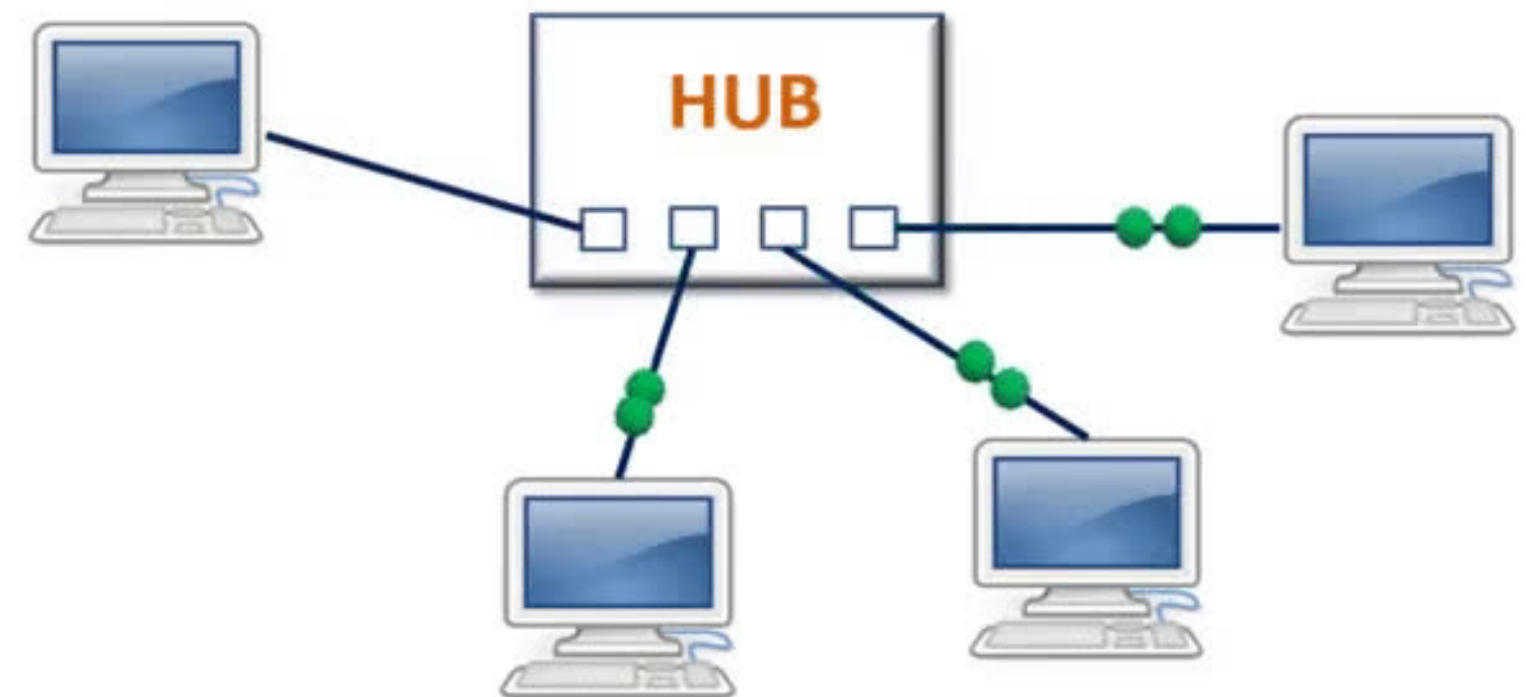


Access Point

Hubs in 1980s

- Function: Broadcasts data to all connected devices
- Doesn't recognize hardware addresses (Mac Addresses)
- Operates at the Physical Layer (Layer 1)

Inefficiency: There is issue with hubs, where data is broadcasted to all devices even when only one device needs it, leading to inefficiency.

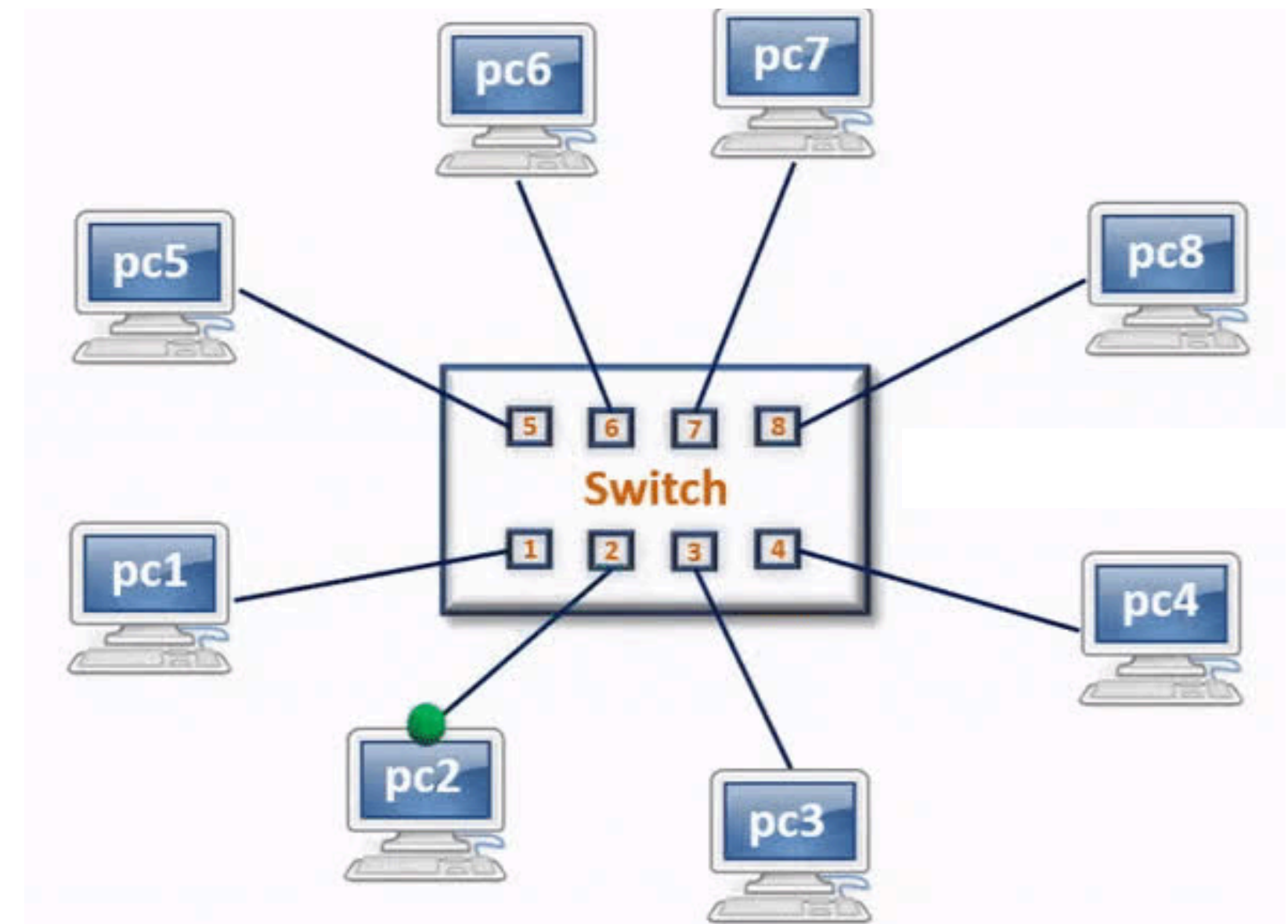


What's the main real-world problem with using a hub in a network with many devices?

Unnecessary Traffic

Switches since 1990s

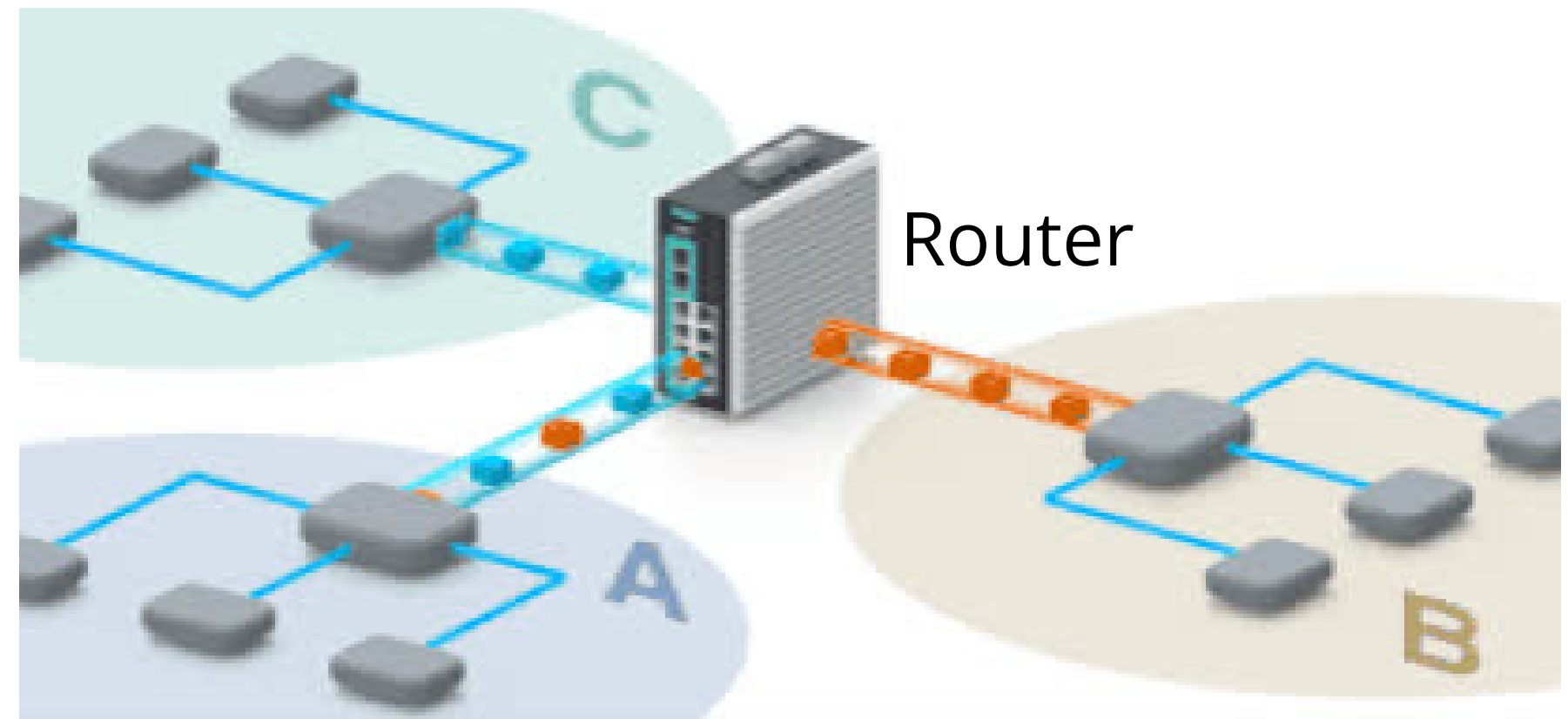
- Operates at the Data Link Layer (Layer 2).
- Connects devices within a LAN and forwards data to the correct device using MAC addresses.
- Has no understanding of layer 3.
- Ports can be configured for connection routing.
- Broadcasting could be controlled using the Mac routing intelligence.



**What if we want to communicate with a
computer in another network**

Router

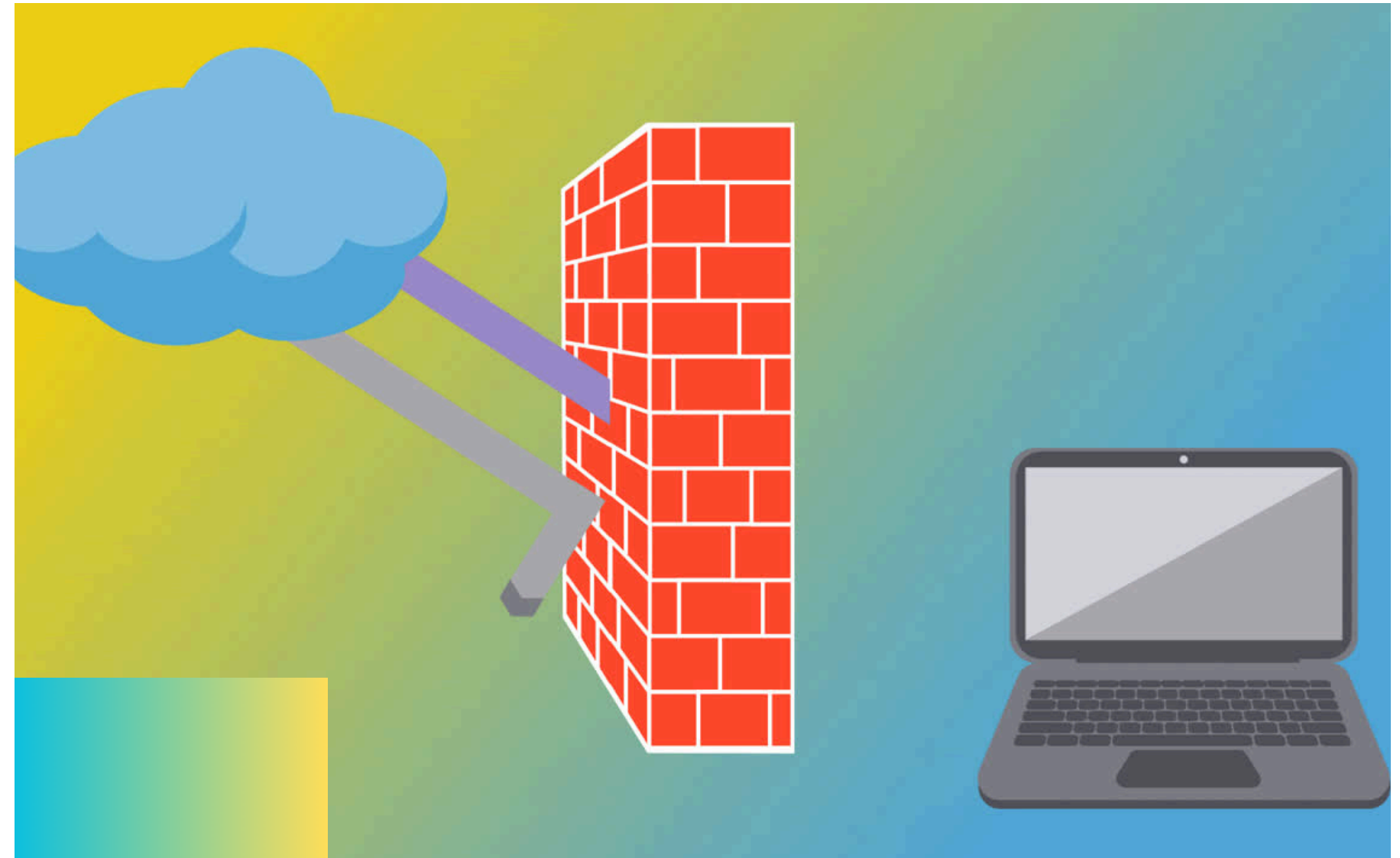
- Routes data between different networks
- Operates at the Network Layer (Layer 3)



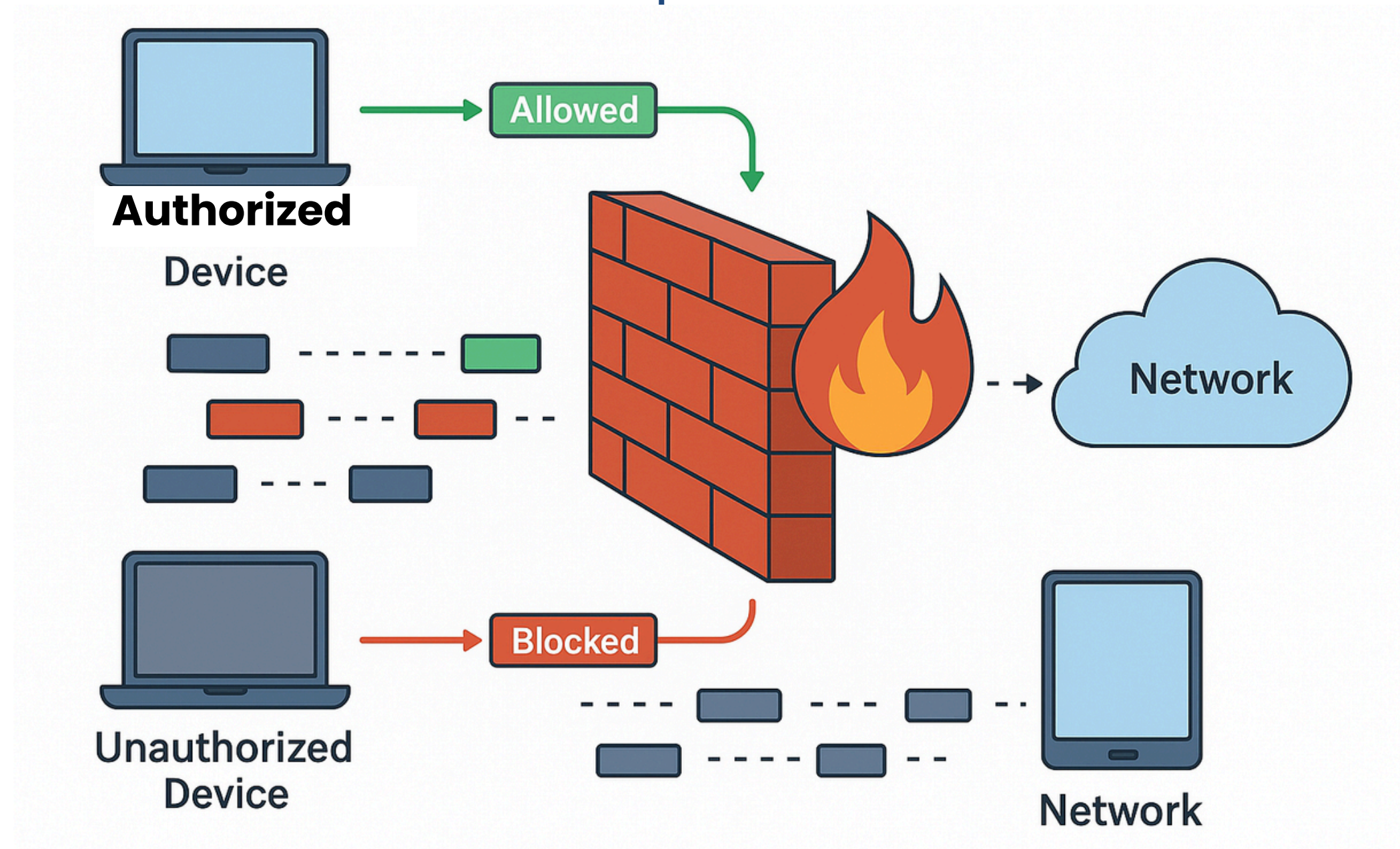
**A router knows where to send packets, but
should every packet be allowed to enter**

Firewall

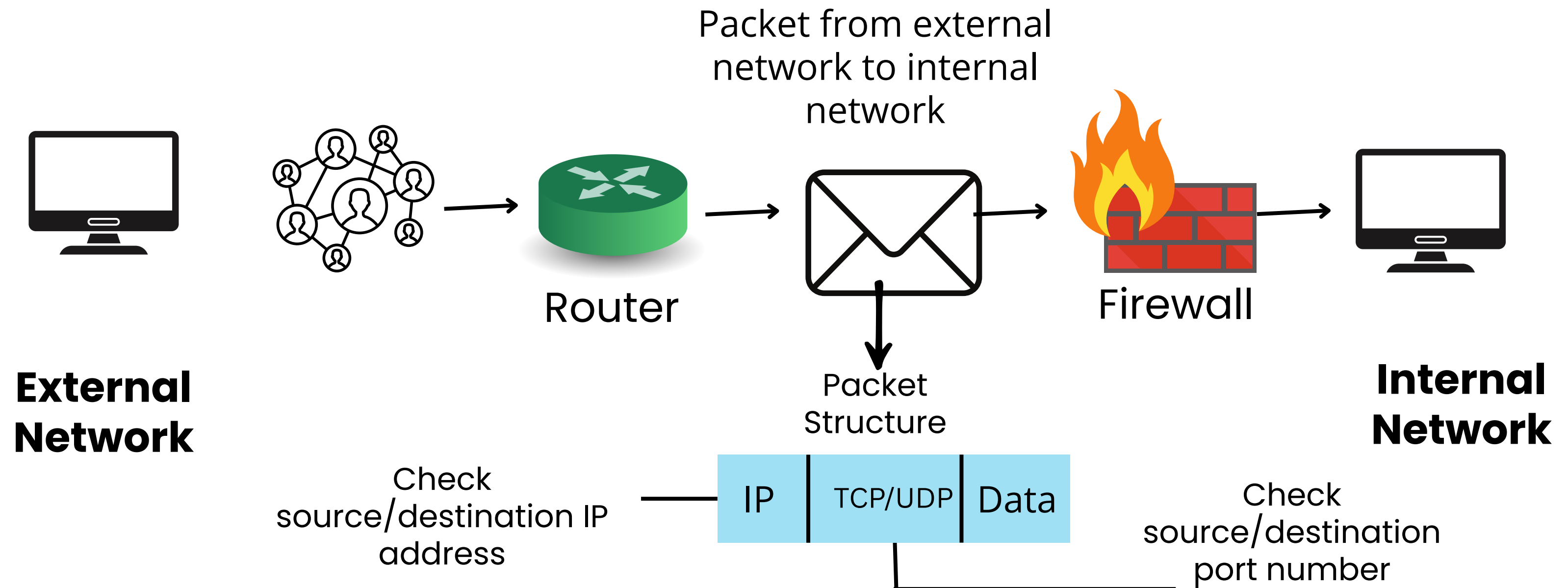
- A firewall is a network security device.
- It monitors data coming in and going out of the network.
- Its main job is to protect the network from unauthorized access or threats.



Firewalls allow only authorized data



What is Packet Filtering?



Types of Firewall

How firewalls remember (or forget) connections

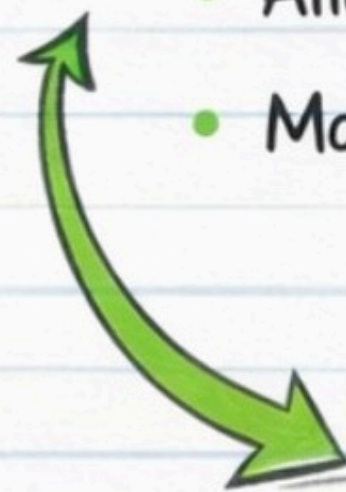
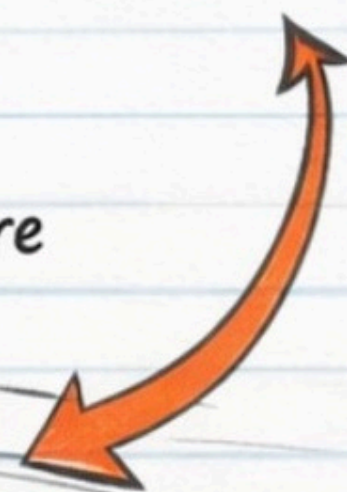
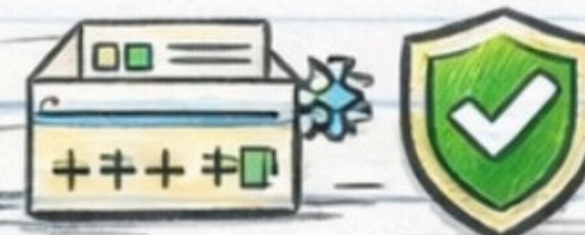
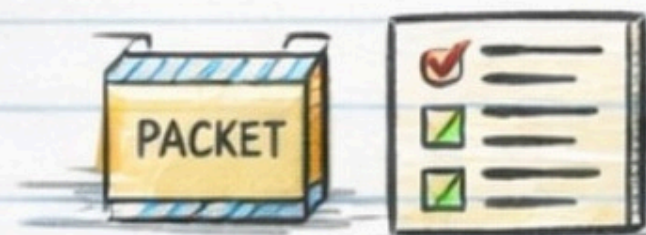
STATELESS FIREWALL

- Checks each packet independently
- No memory of past traffic
- Only uses rules
- Faster but less secure

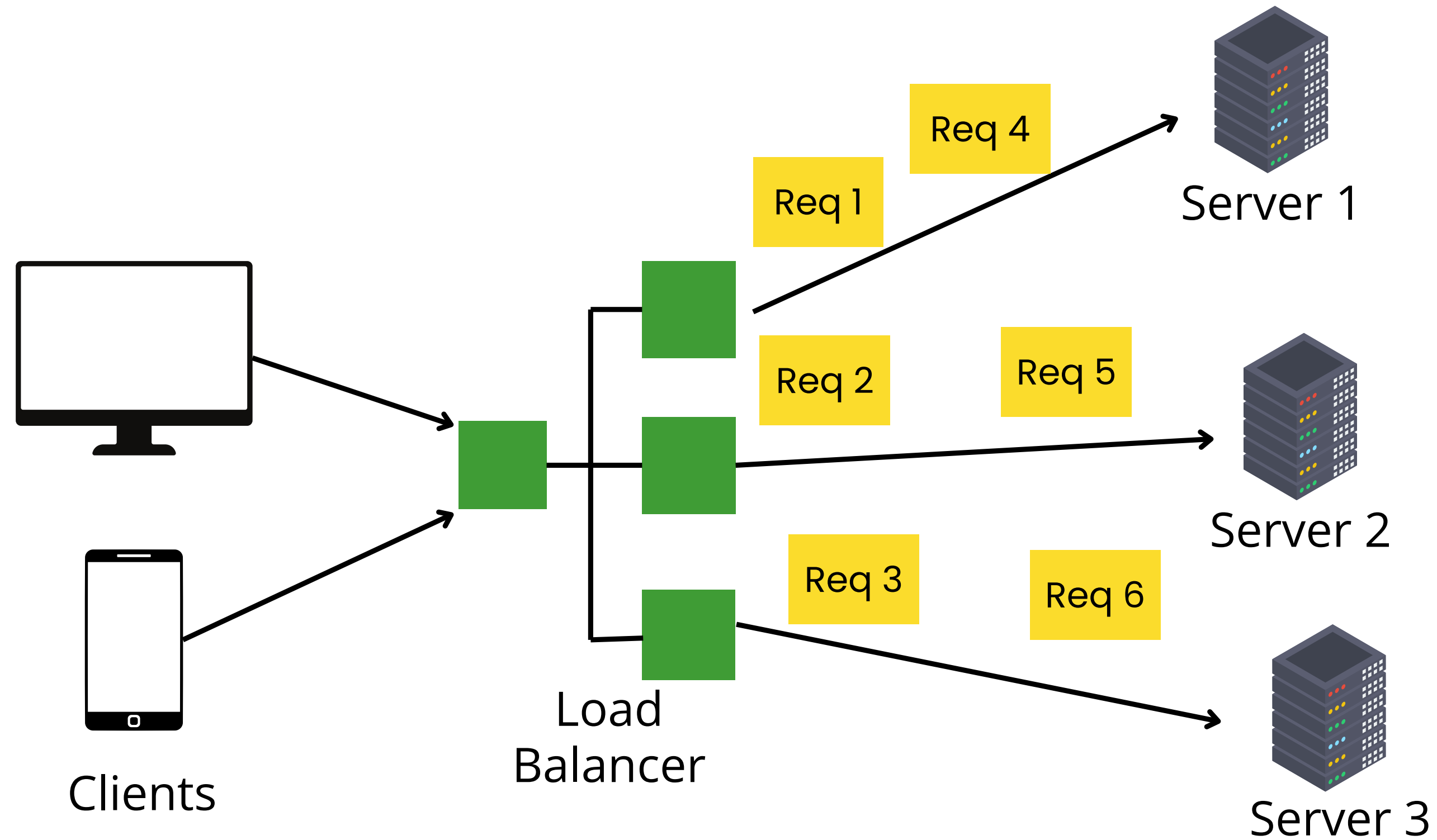


STATEFUL FIREWALL

- Tracks active connections
- Understands sessions
- Allows related packets
- More secure



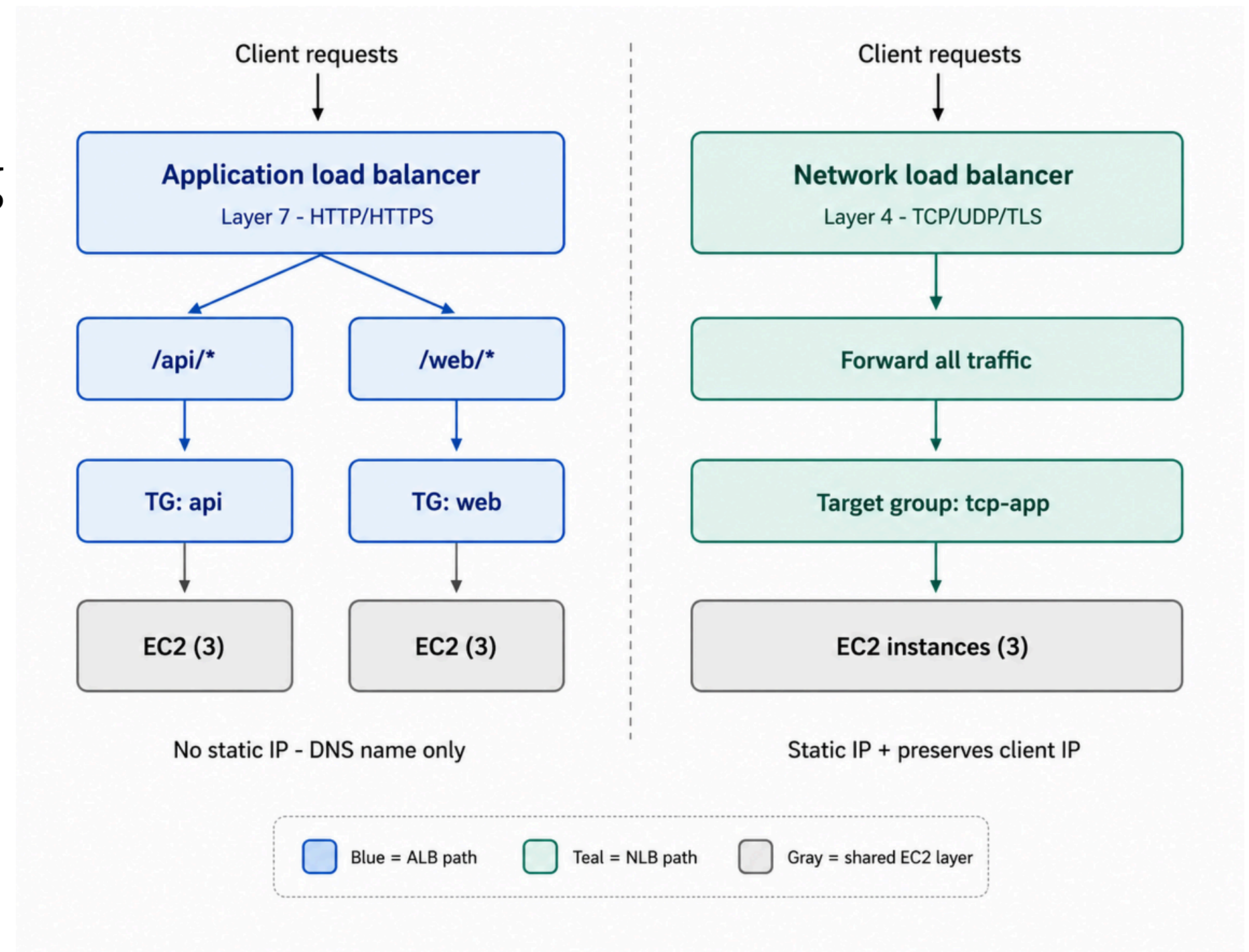
Load Balancers



Load Balancers

A load balancer distributes incoming network traffic across multiple servers to prevent any single server from being overwhelmed.

Major companies like Netflix, Google, Amazon, and Facebook use load balancers to maintain fast and smooth operations, even with high user volumes.



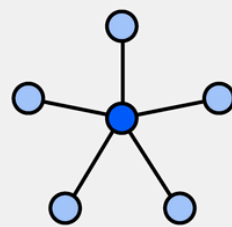
Devices Are Ready... But How Should We Connect Them

Network Topologies

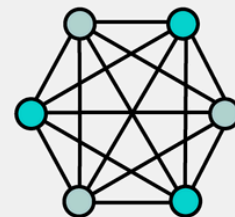
Network topology describes the arrangement of devices and their connections within a network, including how data flows.

Types of Network Topology

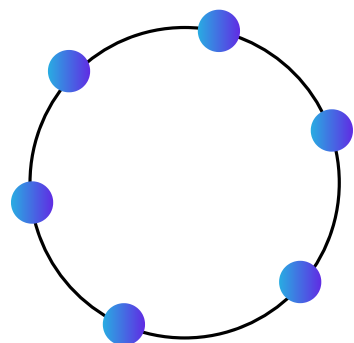
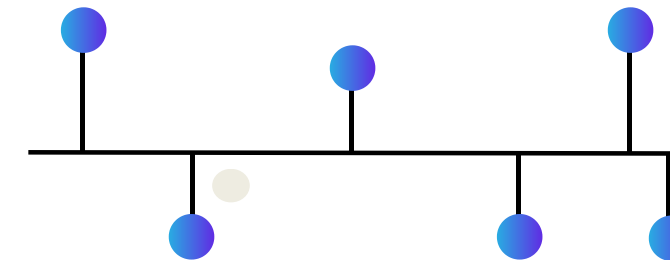
Star



Mesh



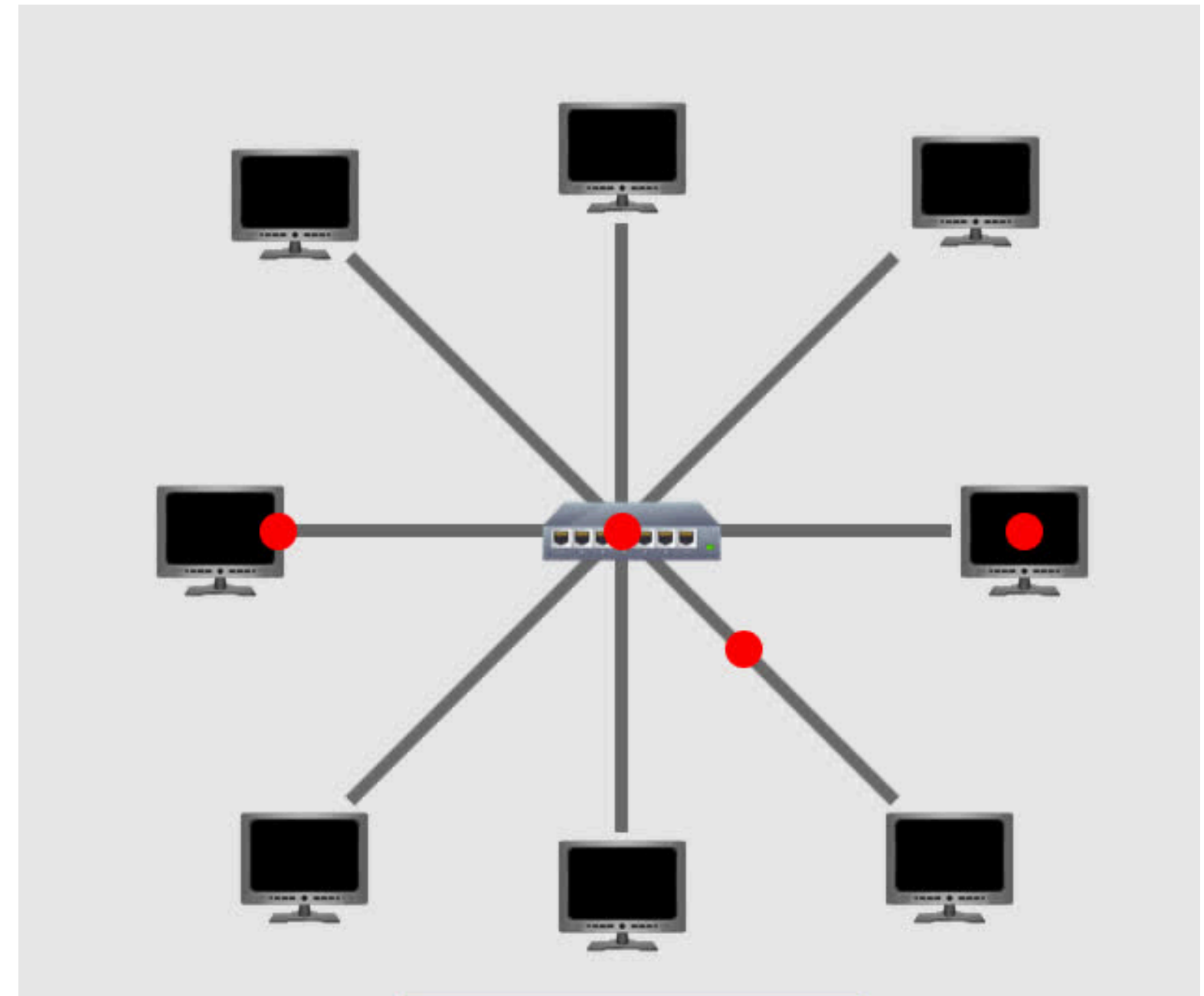
Bus
Topology



Ring
Topology

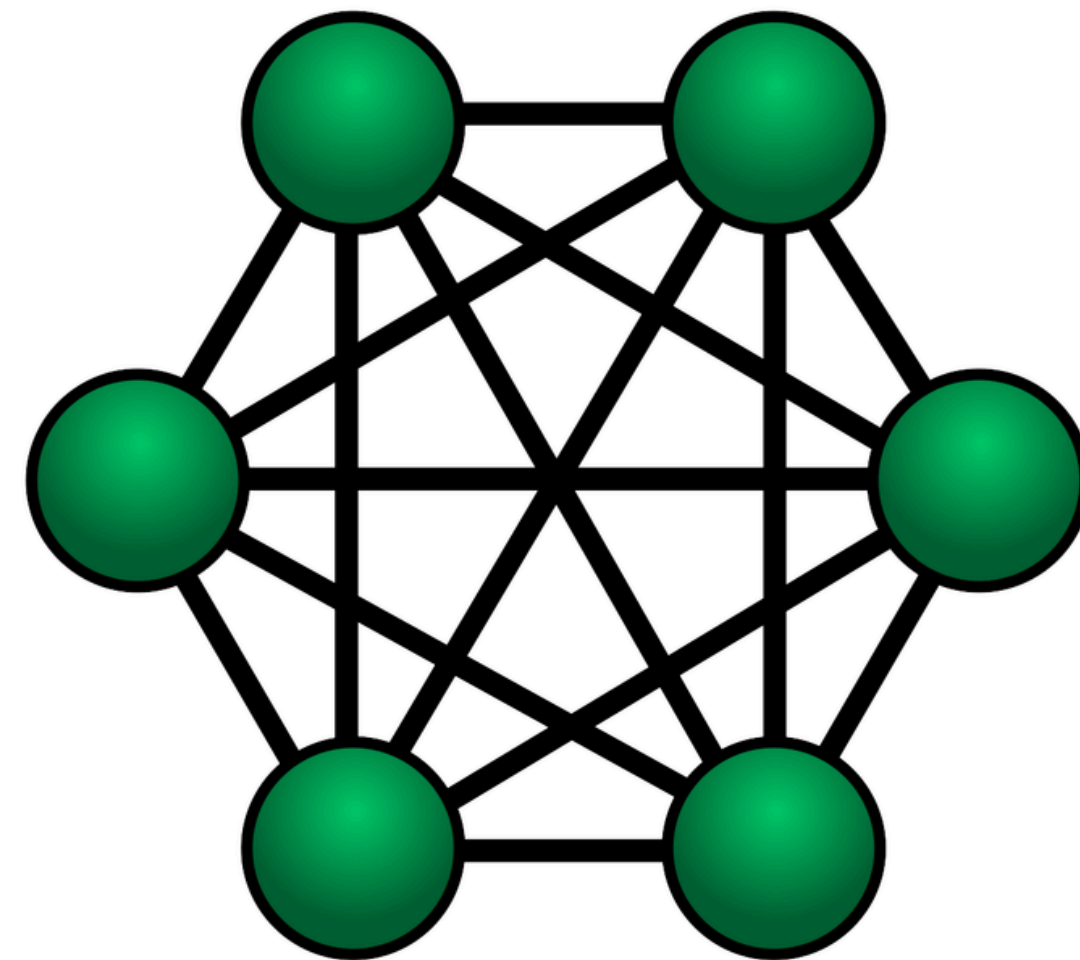
Star topology

- All devices are connected to a central hub or switch.
- Data flows from sender device to the hub/switch, which then sends it to the intended receiver.
- Centralized structure simplifies network diagnostics and maintenance
- Popular choice for Local Area Networks (LANs) due to manageable design



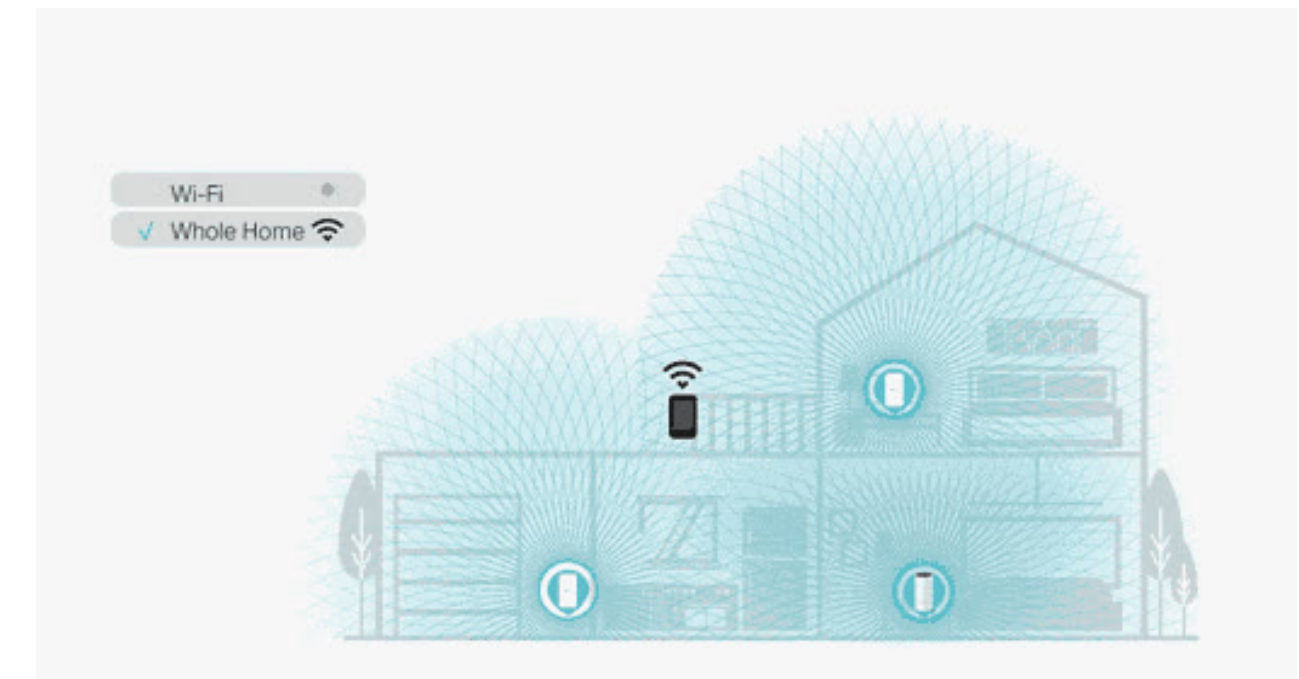
Mesh topology

- In mesh topology, every network device is directly connected to all other devices.
- This creates many paths for sending data between devices.
- It increased reliability because if one path fails, data can take another route.



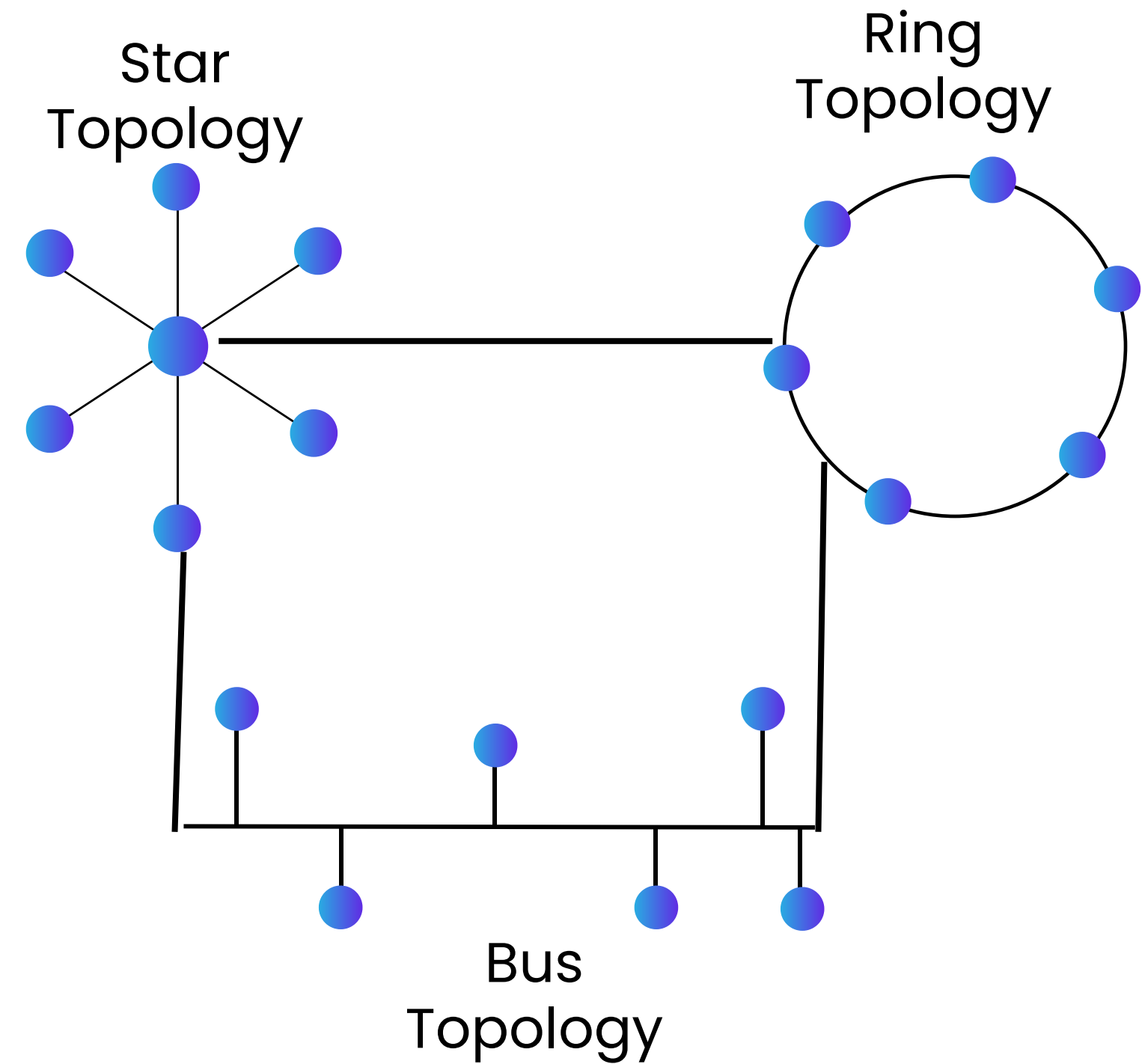
Jokingly called spaghetti networks!

Wireless mesh networks help with WiFi coverage



Hybrid topology

Hybrid topology is a combination of two or more different types of topologies to form a more complex and flexible network.



Environment	Suitable Topology	Reason	Example connected devices
Small Office	Star	Easy setup, centralized control	Laptops, Printers, Routers
Smart Home	Mesh	Reliable, direct device communication	Smart bulbs, cameras
Data Center	Hybrid	Combines benefits of multiple topologies	Servers, Storage Units

Fault Tolerance and Scalability of Topologies

Topology	Fault Tolerance	Scalability
STAR	Medium	Good
MESH	High	Excellent
HYBRID	High	Hard

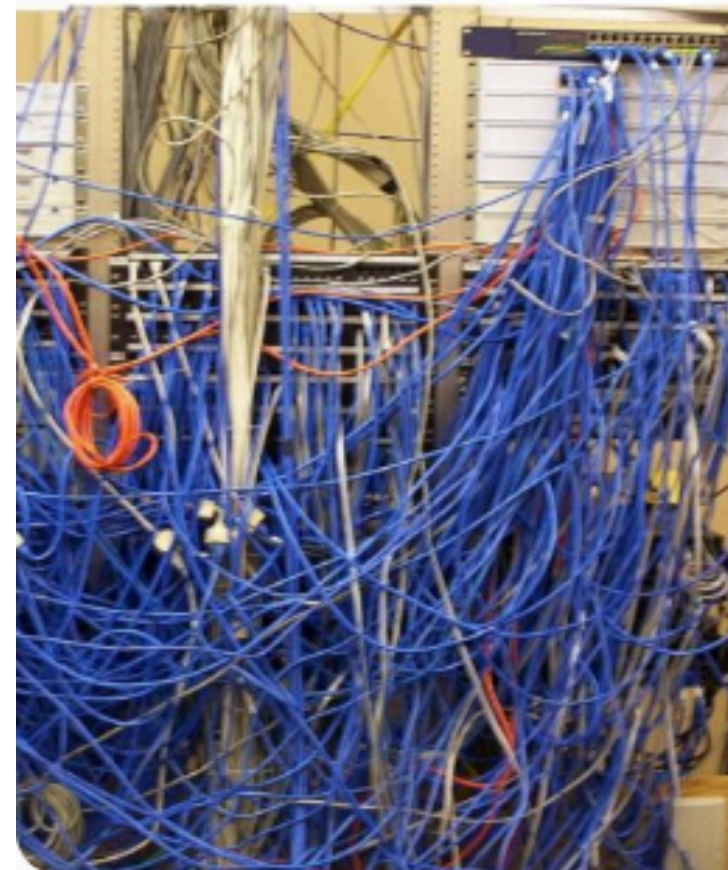
Hardware Meets Software

The Evolutionary Shift

Traditional networking relies on physical boxes, specialized cables, and manual configuration. In the modern cloud era, the **logic remains the same**, but the execution is purely software-defined.

- 🔧 Physical Box → Virtual Instance
- 🔗 Copper Cables → Virtualized Overlays
- Manual Wiring → API / Console Clicks

How it started.



How it's going.



The cloud network

Global Cloud

AWS Foundation

Regions: Isolated geographic areas (e.g., Mumbai) containing multiple data center clusters.

Edge Locations: CDN endpoints that cache content physically closer to the user to reduce latency.

- 🌐 30+ Global Regions
- ⚡ Ultra-low latency backbone



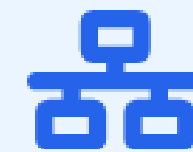
Redundancy: Availability Zones



Isolated Fault Domains

An AZ consists of one or more discrete data centers with redundant power, networking, and connectivity.

Deploying across **two or more AZs** ensures your application survives a fire, flood, or power grid failure at one site.



The Multi-AZ Project

This semester, you will design architectures that split workloads across AZ-A and AZ-B.

Metric: 99.9% availability is a target for high-availability systems.

The Virtual Private Cloud (VPC)

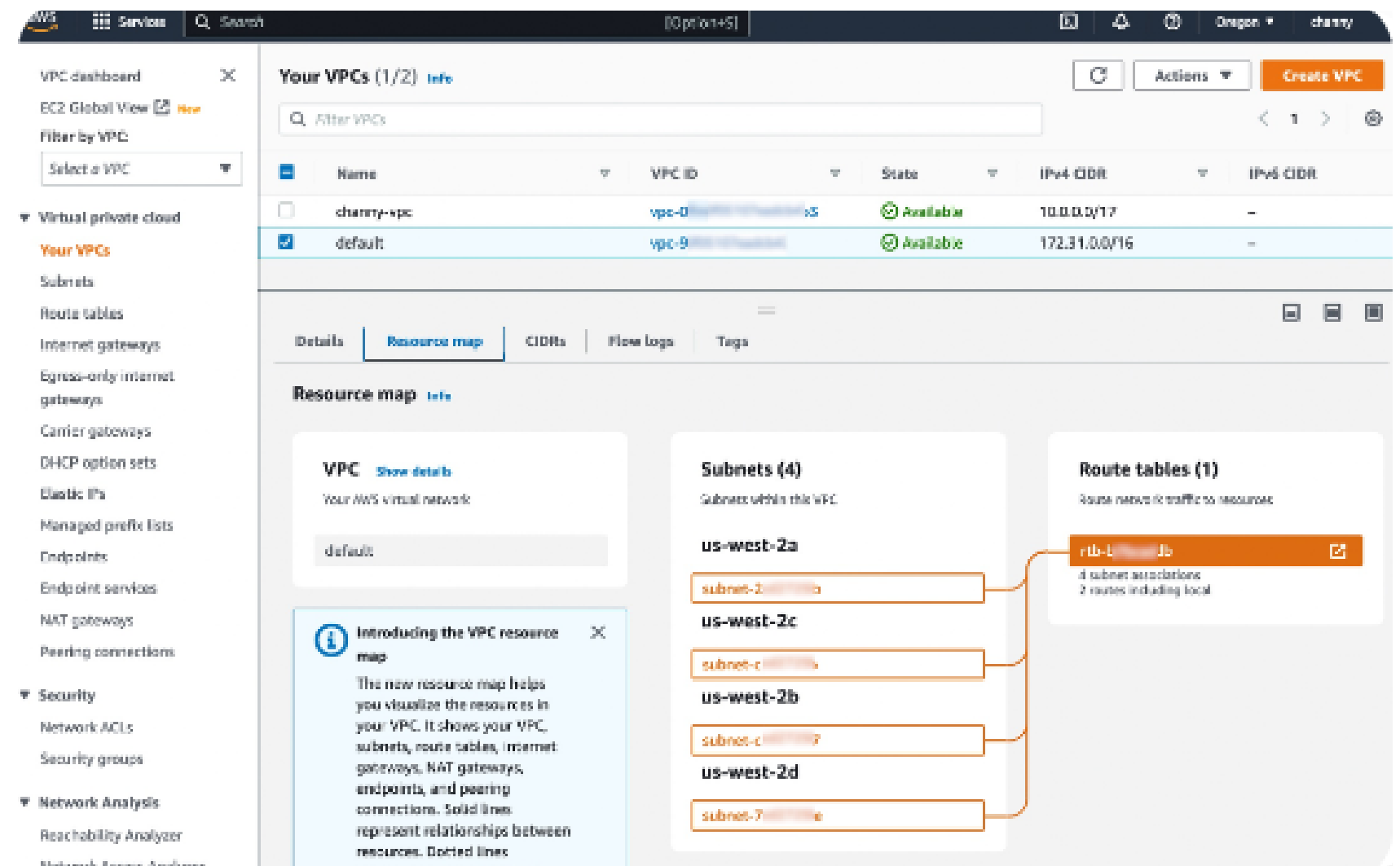
Logical Isolation

A VPC is your private, software-defined network in the AWS cloud. You control IP address ranges, subnets, and routing.

CIDR Addressing Example:

10.0.0.0 / 16 → 65,536 Addresses

Subnets map to specific AZs, partitioning your L3 network for scale and security.



Please fill the feedback form.

**Thanks
for
watching!**